

Exercise 1:

1. Prove that the following functions $\mathbb{R}^n \rightarrow \mathbb{R}$ are convex :
 - (a) $x \mapsto \|x\|$ (any norm)
 - (b) $x \mapsto \max_{i=1,\dots,n} x_i$
 - (c) $x \mapsto \log(e^{x_1} + \dots + e^{x_n})$
2. Prove that the following functions are concave :
 - (a) $\mathbb{R}^n \rightarrow \mathbb{R}, x \mapsto (\prod_{i=1}^n x_i)^{1/n}$ (geometric mean)
 - (b) $\mathbb{S}_{++}^n \rightarrow \mathbb{R}, X \mapsto \log \det X$

Exercise 2: For each of the following functions determine whether it is convex, concave, quasi-convex, or quasiconcave.

1. $f(x) = e^x - 1$ on $\mathbb{R}$.
2. $f(x_1, x_2) = x_1 x_2$ on $\mathbb{R}_{++}^2$.
3. $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbb{R}_{++}^2$.
4. $f(x_1, x_2) = x_1/x_2$ on $\mathbb{R}_{++}^2$.
5. $f(x_1, x_2) = x_1^2/x_2$ on $\mathbb{R}_{++}^2$.
6. $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, on $\mathbb{R}_{++}^2$, where $0 \leq \alpha \leq 1$.

Exercise 3: For a differentiable objective function f_0 , using only the optimality condition : $\forall x$ feasible, $\nabla f_0(x^*)^\top (x - x^*) \geq 0$, derive optimality conditions in the following cases :

1. Unconstrained minimization : $\min. f_0(x)$.
2. Equality constrained minimization : $\min. f_0(x) \quad \text{s.t. } Ax = b$.
3. Minimization over nonnegative orthant : $\min. f_0(x) \quad \text{s.t. } x \succeq 0$.

Exercise 4: Consider the optimization problem

$$\begin{aligned}
 &\min. f_0(x_1, x_2) \\
 &\text{s.t. } 2x_1 + x_2 \geq 1 \\
 &\quad x_1 + 3x_2 \geq 1 \\
 &\quad x_1 \geq 0, \quad x_2 \geq 0.
 \end{aligned}$$

Make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value.

1. $f_0(x_1, x_2) = x_1 + x_2$.
2. $f_0(x_1, x_2) = -x_1 - x_2$.
3. $f_0(x_1, x_2) = x_1$.
4. $f_0(x_1, x_2) = \max\{x_1, x_2\}$.
5. $f_0(x_1, x_2) = x_1^2 + 9x_2^2$.

Exercise 5: Prove that $x^* = (1, 1/2, -1)$ is optimal for the optimization problem :

$$\begin{aligned}
 &\min. \frac{1}{2}x^\top Px + q^\top x + r \\
 &\text{s.t. } -1 \leq x_i \leq 1, \quad i = 1, 2, 3,
 \end{aligned}$$

where :

$$P = \begin{pmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{pmatrix}, \quad q = \begin{pmatrix} -22.0 \\ -14.5 \\ 13.0 \end{pmatrix}, \quad r = 1.$$

Exercise 6: Consider the convex problem

$$\begin{aligned} \min. & f_0(x) \\ \text{s.t.} & f_i(x) \leq 0, \quad i = 1, \dots, m. \end{aligned}$$

Assume x^* and λ^* satisfy the KKT conditions :

$$\begin{aligned} f_i(x^*) &\leq 0, & i = 1, \dots, m \\ \lambda_i^* &\geq 0, & i = 1, \dots, m \\ \lambda_i^* f_i(x^*) &= 0, & i = 1, \dots, m \\ \nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) &= 0. \end{aligned}$$

Show that $\nabla f_0(x^*)^\top (x - x^*) \geq 0$ for all feasible x .

Exercise 7: Solve by hand the following optimization problem :

$$\begin{aligned} \min. & x_1^2 + x_2^2 \\ \text{s.t.} & -2x_1 - x_2 + 10 \leq 0 \\ & -x_1 \leq 0 \end{aligned}$$

Exercise 8: Solve by hand the following optimization problem :

$$\begin{aligned} \min. & 5x_1^2 + 6x_2^2 \\ \text{s.t.} & x_1 - 4 \leq 0 \\ & 25 - x_1^2 - x_2^2 \leq 0 \end{aligned}$$

Exercise 9: Use the Lagrangian conditions to solve the following problem on the domain $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 > 0\}$:

$$\begin{aligned} \min. & x_1 + 2/x_2 \\ \text{s.t.} & -x_2 + 1/2 \leq 0 \\ & -x_1 + x_2^2 \leq 0 \end{aligned}$$

Exercise 10: Consider an optimization problem with feasible set defined by inequalities only :

$$X = \{x \mid f_i(x) \leq 0, i = 1, \dots, p\}.$$

For any point $\bar{x} \in X$, define the active set

$$I(\bar{x}) = \{i \mid f_i(\bar{x}) = 0\}.$$

Let us remind Slater's constraint qualification (SCQ), linear independence constraint qualification (LICQ), Mangasarian-Fromovitz constraint qualification (MFCQ) :

$$(\text{SCQ}) : \exists x_0 \quad f_i(x_0) < 0, \quad i = 1, \dots, p.$$

$$(\text{LICQ}) \text{ at a point } \bar{x} \in X : \{ \nabla f_i(\bar{x}) : i \in I(\bar{x}) \} \text{ linearly independent}$$

$$(\text{MFCQ}) \text{ at a point } \bar{x} \in X : \exists d \quad \langle \nabla f_i(\bar{x}), d \rangle < 0 \quad \text{for all } i \in I(\bar{x})$$

1. Prove that if the f_i are all convex, then (SCQ) implies (MFCQ).
 2. Prove that (LICQ) implies (MFCQ).
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Exercise 11: Let $A \in \mathbb{R}^{p \times n}$.

1. For $p \geq n$ and $\text{rank } A = n$, solve the problem :

$$\min. \|Ax - b\|_2^2$$

2. For $p \leq n$ and $\text{rank } A = p$, consider the problem :

$$\begin{aligned} \min. & \frac{1}{2} \|x\|_2^2 \\ \text{s.t.} & Ax = b \end{aligned}$$

- (a) Write the Lagrangian $L(x, \nu)$, derive the dual function $g(\nu)$ and the dual problem.
 - (b) Give Slater's sufficient condition for strong duality and solve the dual problem.
 - (c) Write the KKT conditions.
 - (d) Solve the primal and find x^* .
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Exercise 12: constraint qualification Consider the optimization problem :

$$\begin{cases} \min_{x \in \mathbb{R}^2} x_1^2 + x_2^2 \\ \text{s.t. } (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ \quad (x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \end{cases}$$

1. Make a sketch of the problem. What is the optimal point x^* ? Are Slater's qualification constraints satisfied?
 2. Write the Lagrangian and the KKT optimality conditions. Are there Lagrange parameters proving optimality of x^* ?
 3. Write and solve the dual. Does strong duality hold? What is the optimum value for the dual?
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Exercise 13: failure of KKT Consider the problem on $\mathbb{R}^2$:

$$\begin{cases} \min. (x_1 + 1)^2 + x_2^2 \\ \text{s.t. } -x_1^3 + x_2^2 \leq 0 \end{cases}$$

1. Sketch the feasible set and solve the problem.
 2. Find multipliers λ_0, λ_1 satisfying the Fritz-John conditions.
 3. Prove there exist no Lagrange multiplier vector for the optimal solution. Explain why not.
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Exercise 14: subdifferential Prove the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are convex and calculate ∂f :

1. $f(x) = |x|$,
 2. $f(x) = \iota_{\mathbb{R}_+}(x)$
 3. $f(x) = \begin{cases} -\sqrt{x} & \text{if } x \geq 0, \\ +\infty & \text{otherwise.} \end{cases}$
 4. $f(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x = 0, \\ +\infty & \text{otherwise.} \end{cases}$
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Exercise 15: For any norm $\|\cdot\|$ on $\mathbb{R}^n$, let us recall that the dual norm is defined by $\|z\|_* := \sup_{\|x\| \leq 1} \langle z, x \rangle$ (where $\langle u, v \rangle = u^\top v$ is the standard inner product).

1. Show that the ℓ_2 norm $\|x\|_2 = \sqrt{x^\top x}$ is self-dual.
 2. Let $P \in \mathbb{S}_{++}^n$ be a positive definite matrix and $\|x\|_P = \sqrt{x^\top P x}$. Compute the dual norm $\|\cdot\|_{P^*}$.
 3. Show that norms $\|x\|_\infty := \max_{i=1}^n |x_i|$ and $\|x\|_1 = \sum_{i=1}^n |x_i|$ are dual of each other.
 4. Show that, if $p, q > 1$ are such that $\frac{1}{p} + \frac{1}{q} = 1$, then the norms $\|x\|_p = (\sum_{i=1}^n x_i^p)^{1/p}$ and $\|x\|_q = (\sum_{i=1}^n x_i^q)^{1/q}$ are dual of each other.
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Exercise 16: Derive the Fenchel conjugates of the following functions.

1. affine function : $f(x) = ax + b$.
 2. exponential : $f(x) = e^x$.
 3. negative logarithm : $f(x) = -\log x$.
 4. quadratic function : $f(x) = \frac{1}{2}x^\top Q x$ with $Q \in \mathbb{S}_{++}^n$.
 5. square norm : $f(x) = \frac{1}{2}\|x\|_2^2$
 6. norm : $f(x) = \|x\|$.
 7. $f(x) = x \log x$ for $x \geq 0$ and $+\infty$ otherwise.
 8. $f(x) = 1/x$ for $x > 0$ and $+\infty$ otherwise.
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Exercise 17: Derive the Fenchel conjugate and biconjugate of the following functions ($a > 0$ and all function are $\mathbb{R} \rightarrow \mathbb{R}$).

1. $f(x) = a|x|$.
 2. $f(x) = \iota_{[-a, a]}(x)$.
 3. $f(x) = +\infty$ for $x > a$ and $f(x) = 0$ for $x \leq a$.
 4. $f(x) = \iota_{\{0\}}(x)$.
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Exercise 18: Farkas' lemma Consider the following linear program (LP) with variable $x \in \mathbb{R}^n$ ($c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ are given) :

$$p^* : \begin{cases} \min. & c^\top x \\ \text{s.t.} & Ax = b \\ & x \succeq 0 \end{cases}$$

1. Write the Lagrangian and give the Lagrange dual function.
2. Write the dual problem.

3. Consider the case $c = 0$ and prove that exactly one of the assertions below holds, but not both :

$$\text{Either : } \exists x \in \mathbb{R}^n, Ax = b, x \succeq 0 \quad \text{or : } \exists y \in \mathbb{R}^m, A^\top y \succeq 0, b^\top y < 0$$

Exercise 19: Lagrange and Fenchel duality for LPs

1. Let $A \in \mathbb{R}^{m \times n}$, $c \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. Consider the following (primal) LP with variable $x \in \mathbb{R}^n$:

$$\begin{aligned} \min. & c^\top x \\ \text{s.t.} & Ax = b \\ & x \succeq 0 \end{aligned}$$

Find the dual problem using Lagrange duality. We suggest to denote y (resp. s) the vector of dual variables associated to equality (resp. inequality) constraints; note that s can be eliminated.

2. Write the KKT optimality conditions associated to the above LP optimization problem.
3. Define

$$f(x) = \begin{cases} c^\top x & \text{if } x \succeq 0, \\ +\infty & \text{otherwise.} \end{cases} \quad \iota_{\{b\}}(x) = \begin{cases} 0 & \text{if } x = b, \\ +\infty & \text{otherwise.} \end{cases}$$

Derive the Fenchel conjugates of both functions.

4. Derive the Fenchel dual of the problem $\min f(x) + \iota_{\{b\}}(Ax)$ and show that the Lagrange and Fenchel dual of the primal LP are the same.
5. Consider now the problem

$$\begin{cases} \min. & c^\top x - \mu \sum_{i=1}^n \log x_i \\ \text{s.t.} & Ax = b \end{cases}$$

Write the Lagrangian and the KKT optimality conditions. Show that the above problem can be considered as a perturbation of the previous problem.

Exercise 20: Consider the problem

$$\min. \|x\|_1 \quad \text{s.t. } Ax = b.$$

We write $A = [a_1, \dots, a_n]$ where $(a_i)_{i=1, \dots, n}$ are the columns of A .

1. Defining $x^+, x^- \succeq 0$ and decomposing $x = x^+ - x^-$, write an equivalent problem.
2. Write the Lagrangian, derive the Lagrange dual function and the dual problem.
3. Write the KKT optimality conditions. Show that the optimal x has a nonzero i th component x_i only if $|a_i^\top u| = 1$.
4. Determine the Fenchel conjugates of $f(x) = \|x\|_1$ and $g(y) = \iota_b(y)$.
5. Write the above optimization problem using the previous two functions and determine its Fenchel dual problem.

Exercise 21:

1. Let $f(x) = \frac{1}{2}\|x\|_2^2$. Derive the Fenchel conjugate of f .
2. Let f be a function such that $f^* = f$, where f^* is the Fenchel conjugate of f .

- (a) Write the Fenchel-Young inequality and show that necessarily $f(y) \geq \frac{1}{2}\|y\|_2^2$ for all y .
- (b) Deduce that $f^*(y) \leq \frac{1}{2}\|y\|_2^2$.
3. Conclude about sufficient and necessary conditions for a function to be equal to its Fenchel conjugate.

Exercise 22: Fenchel weak/strong duality Let $f : \mathbb{R}^n \rightarrow]-\infty, +\infty]$ and $g : \mathbb{R}^m \rightarrow]-\infty, +\infty]$ be l.s.c. convex proper functions. Let $A \in \mathbb{R}^{m \times n}$ be a matrix. Consider the following primal optimization problem with value p^* and its associated dual problem with value d^* :

$$(p^*) : \min_{x \in \mathbb{R}^n} f(x) + g(Ax) \quad (d^*) : \max_{v \in \mathbb{R}^m} -f^*(-A^\top v) - g^*(v)$$

1. Show that $d^* \leq p^*$ (weak duality).
2. Suppose $d^* = p^*$ (strong duality). Show that x^*, v^* are primal, dual optimal if and only if :

$$-A^\top v^* \in \partial f(x^*) \quad \text{and} \quad v^* \in \partial g(Ax^*)$$

Exercise 23: penalized linear regression Consider the following optimization problem, where $\lambda > 0$, $b \in \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times n}$ are fixed :

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}\|Ax - b\|_2^2 + \frac{\lambda}{2}\|x\|_2^2$$

1. Find the analytic optimal solution x^* .
2. Note that the above problem can be written :

$$\begin{aligned} \min_{x \in \mathbb{R}^n, \xi \in \mathbb{R}^m} \quad & \frac{1}{2}\|\xi\|_2^2 + \frac{\lambda}{2}\|x\|_2^2 \\ \text{s.t.} \quad & \xi = Ax - b \end{aligned}$$

- (a) Write the Lagrangian of the above constrained problem (where α denotes the vector of dual variables).
- (b) Write the KKT optimality conditions of the above constrained problem.
- (c) Solve the KKT conditions. Give first the optimal dual variable α^* , then derive the optimal primal variable x^* .
- (d) Compare with the original analytic solution.
3. We consider now the Fenchel dual problem of the original problem.
 - (a) Write the Fenchel dual. We suggest to write the problem in the form $\min_{x \in \mathbb{R}^n} f(x) + g(Ax)$ and recall properties of the Fenchel conjugate (scaled/translated function, case of $\frac{1}{2}\|\cdot\|_2^2$).
 - (b) Write the subdifferential conditions for (x^*, α^*) to be primal/dual optimal. Show that they correspond to the KKT conditions.

Exercise 24: For the following functions, derive the proximal operator $\text{prox}_{\lambda f}$ (where $\lambda > 0$) :

1. square loss : $f(x) = \frac{1}{2}x^2$.
2. absolute value : $f(x) = |x|$.
3. support function : $f(x) = \sigma_{[a,b]}(x) = \begin{cases} ax & \text{if } x \leq 0 \\ bx & \text{if } x \geq 0 \end{cases}$ with $a \leq b$.

Note that $\sigma_{[a,b]} = \iota_{[a,b]}^*$ is known as the support function of the set $[a, b]$.

4. ReLU function : $f(x) = \sigma_{[0,1]}(x) = \max(x, 0)$.

5. dead-zone linear / ω -insensitive loss : $f(x) = \max(0, |x| - \omega)$.
 6. Huber : $f(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \leq \omega \\ \omega|x| - \frac{\omega^2}{2} & \text{otherwise.} \end{cases}$
 7. Log : $f(x) = -\log x$ if $x > 0$ and $+\infty$ otherwise.
 8. Quadratic : $f(x) = \frac{1}{2}x^\top Ax + b^\top x + c$ with $A \succeq 0$ and symmetric.
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Exercise 25: Let $m \leq n$ be two integers, $A \in \mathbb{R}^{m \times n}$ a rank m matrix and $b \in \mathbb{R}^m$. Let $c \in \mathbb{R}^n$.

1. Consider the problem :

$$\begin{aligned} \min_x \quad & \frac{1}{2} \|x - c\|^2 \\ \text{s.t.} \quad & Ax = b. \end{aligned}$$

- (a) Write the KKT conditions.
 - (b) Solve the KKT equations and find the optimal point.
2. Define $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ by $f(x) = 0$ if $Ax = b$ and $f(x) = +\infty$ if $Ax \neq b$. Determine $\text{prox}_f(c)$.